



INFINITE SERIES



SEQUENCES

A sequence is a set of real numbers $a_1, a_2, a_3, a_4, \dots, a_n$ in a particular order or a set of numbers which follows a pattern & is denoted by a_n

Ex: $\left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n}\right)$

If the numbers of terms in a given sequence is unlimited, then the given sequence is said to be an infinite sequence.

Ex: $(1, 2, 3, 4, \dots)$



CONVERGENCE, DIVERGENCE, & OSCILLATION:

Given a sequence $a_1, a_2, a_3, a_4, \dots, a_n, \dots$

Let $\lim_{n \rightarrow \infty} a_n = l$ Where;

- i) l is finite & unique then the sequence is said to be a convergent sequence.
- ii) l is not finite then the sequence is said to be a divergent sequence.
- iii) l is finite but not unique then the given sequence is said to be a oscillatory sequence.



EXAMPLES:

1) 1,2,3,4,...

Soln: Here $a_n = n$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} n = \infty$$

Given sequence is Divergent.



$$2) a_n = \frac{n^2 - 2n}{3n^2 + n}$$

$$\text{soln: } \lim_{n \rightarrow \infty} \frac{n^2 - 2n}{3n^2 + n} = \frac{1}{3}$$

Given sequence is convergent.



$$3) a_n = 3 + (-1)^n$$

$$\begin{aligned} \text{Soln: } \lim_{n \rightarrow \infty} a_n &= \lim_{n \rightarrow \infty} (3 + (-1)^n) = 4; n \text{ is even.} \\ &= 2; n \text{ is odd.} \end{aligned}$$

Given sequence is oscillatory.



SERIES:

A series is formed by the sum of the terms of a sequence.

Ex: $1+3+5+7+\dots+(2n-1)$.

If the numbers of terms in a series is unlimited then it is called as an infinite series.

Infinite series is denoted by $a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$ or $\sum_{n \rightarrow \infty} a_n$ or simply $\sum a_n$.



CONVERGENCE, DIVERGENCE, & OSCILLATION:

Consider the infinite series, $\sum u_n = u_1 + u_2 + u_3 + \dots + u_n + \dots$ & let the sum of the first n terms be $s_n = u_1 + u_2 + u_3 + \dots + u_n$.

Clearly s_n is a function of n & as n increases indefinitely three possibility arises:

- i) If s_n tends to a finite limit as $n \rightarrow \infty$, the series $\sum u_n$ is said to be a convergent series.
- ii) If s_n tends to $\pm\infty$ as $n \rightarrow \infty$, the series $\sum u_n$ is said to be a divergent series.
- iii) If s_n does not tend to unique limit as $n \rightarrow \infty$, then the series $\sum u_n$ is said to be a oscillatory series.



EXAMPLES:

1) $1+2+3+4+\dots$

Soln: Here $s_n = 1+2+3+\dots+n = \frac{n(n+1)}{2}$

next, $\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \frac{n(n+1)}{2} \rightarrow \infty.$

Given series is divergent.



$$2) 5-4-1+5-4-1+5-4-1+\dots$$

Soln: Here $s_n = 5-4-1+5-4-1+\dots+n$ term

$= 0, 5, \text{ or } 1$

Given series is oscillatory.



$$3) 1+4+9+16+\dots=\sum_{r=1}^{\infty} r^2$$

$$\text{soln: } S_n = \sum_{r=1}^n r^2$$

$$= \frac{n(n+1)(2n+1)}{6}$$

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \left(\frac{n(n+1)(2n+1)}{6} \right)$$

$$= \infty$$

Given series is divergent.



GEOMETRIC SERIES:

The series $1 + r + r^2 + r^3 + \dots \infty$

- i) Converges if $|r| < 1$
- ii) Diverges if $r \geq 1$
- iii) Oscillates if $r \leq -1$.